

Lecture Scribe Exam Prep



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PDF

I want you to generate a Lecture Scribe that can be used for Exam Preparation. Make sure to use only the lecture slides shared with you. List of things that scribe should have are: Definitions and notation, Assumptions and conditions, Statements of theorems or results, Step by step proofs, Worked examples with all intermediate steps included. Everything should be from the lecture slides. Include step by step reasoning within the final scribe for maximum clarity for revision. Do not introduce any external intuition, creative explanations, or examples not found in the provided materials. Remember that this lecture scribe is for revision purpose of exam. Important: I will use the same prompt and the same context materials for multiple independent runs to generate different outputs. After that I will select the best scribe that is been given to me.

Below is a **Lecture Scribe for Exam Preparation**, constructed **strictly from the provided lecture slides** of *CSE400 – Lecture 11: Transformation of Random Variables*. All definitions, notation, assumptions, statements, derivations, and examples are written only from the lecture material, including the handwritten derivations and diagrams shown in the slides. No external explanations or additional intuition have been added.

Lecture Scribe — Transformation of Random Variables

Course: CSE400 – Fundamentals of Probability in Computing

Lecture: 11 – Transformation of Random Variables

Source: Lecture Slides by Dhaval Patel, PhD

1. Learning Objectives (From Outline Slide)

The lecture covers:

1. Transformation of random variables
2. Function of two random variables
3. Illustrative example: derivation for the case

$$Z = X + Y$$

The key objective stated in the slides is:

How to find the PDF of a new transformed random variable.

2. Definitions and Notation

Random Variables and Transformations

- Let X be a random variable with known probability density function (PDF)

$$f_X(x)$$

- A transformed random variable is defined as:

$$Y = g(X)$$

- The cumulative distribution function (CDF) of Y is:

$$F_Y(y) = P(Y \leq y)$$

- If X has CDF $F_X(x)$, then transformation proceeds through the CDF relation shown in the slides.
-

Function of Two Random Variables

- Given two random variables X and Y , a new variable may be defined as:

$$Z = g(X, Y)$$

- Example considered in lecture:

$$Z = X + Y$$

3. Assumptions and Conditions

From the slides:

1. The PDF of the original random variable is **known a priori**.

2. The function $Y = g(X)$ is considered under:
 - monotonically increasing case
 - monotonically decreasing case
3. For the two-variable example:
 - X and Y may be independent (as stated in problem statements).
4. Limits of transformation must be changed according to the transformation.

These assumptions appear explicitly in the transformation steps and example slides.

4. Transformation of One Random Variable

Problem Setup

Given:

$$Y = g(X)$$

Goal:

Find $f_Y(y)$.

Step 1 — Express CDF of Transformed Variable

From the slide derivation:

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

For monotonic increasing case:

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Thus,

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2 — Differentiate to Obtain PDF

Differentiate with respect to y :

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using chain rule (as written in slide derivation):

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

or equivalently,

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

Monotonically Decreasing Case

From the slide:

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Differentiation gives:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|_{x=g^{-1}(y)}$$

Final Result (As Boxed in Slide)

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \text{ evaluated at } x = g^{-1}(y)$$

5. Worked Example — Transformation of One RV

Given

- $X \sim \text{Uniform}(-1, 1)$
- Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

From slide:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 1 — Inverse Transformation

From slide derivation:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 2 — Compute Derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 3 — Apply Transformation Formula

From slide computation:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Hence,

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

and 0 otherwise.

6. Function of Two Random Variables

Problem Statement

Let:

$$Z = X + Y$$

Find the PDF of Z .

The slides specify:

1. Find $f_Z(z)$
2. Case when X and Y are independent
3. Example cases include:
 - $X, Y \sim N(0, 1)$
 - exponential distributions with parameter λ

(Only statements are provided in slides; derivation structure is shown graphically.)

7. Derivation for $Z = X + Y$

Step 1 — Start from Definition

From the slide:

$$F_Z(z) = P(Z \leq z)$$

Substitute definition:

$$F_Z(z) = P(X + Y \leq z)$$

Step 2 — Express Using Joint PDF

From graphical derivation in slide:

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x,y) dx dy$$

The shaded region in the slide corresponds to the area satisfying $x + y \leq z$.

Step 3 — Integration Representation

The slide shows integration written in two equivalent ways:

Horizontal strip integration

$$\int \left(\int f_{X,Y}(x,y) dx \right) dy$$

Vertical strip integration

$$\int \left(\int f_{X,Y}(x,y) dy \right) dx$$

Step 4 — Obtain PDF

Differentiate:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

This produces the PDF of Z .

8. Summary of Results (From Slides)

1. Transformation method:
 - Use CDF
 - Apply inverse transformation
 - Differentiate to obtain PDF
2. Final transformation formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

3. For two random variables:
 - Express CDF using joint distribution
 - Integrate over region defined by transformation
 - Differentiate to obtain PDF
-

End of Lecture Scribe — Prepared strictly from Lecture Slides for exam revision.



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Lecture Scribe — Transformation of Random Variables

Course: CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Source: Lecture Slides by Dhaval Patel, PhD

This lecture scribe is written strictly from the provided lecture slides. All notation, assumptions, derivations, and examples follow the structure and content shown in the slides, including the handwritten derivation steps.

1. Definitions and Notation

1.1 Transformation of a Random Variable

Let:

- X be a random variable with known PDF

$$f_X(x)$$

- A new random variable is defined as

$$Y = g(X)$$

Definitions used in the slides:

- CDF of Y :

$$F_Y(y) = P(Y \leq y)$$

- PDF of Y :

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

The objective stated in the outline is:

Learning transformation techniques for random variables and how to find the PDF of the transformed random variable.

1.2 Function of Two Random Variables

Given two random variables X and Y :

- A new random variable may be defined as

$$Z = g(X, Y)$$

Illustrative example considered in the lecture:

$$Z = X + Y$$

2. Assumptions and Conditions

From the lecture slides:

1. The PDF of the original random variable is known a priori.
2. The transformation $Y = g(X)$ is treated under monotonic cases:
 - monotonically increasing
 - monotonically decreasing
3. For functions of two random variables:
 - joint PDF $f_{X,Y}(x, y)$ is used.
4. Limits of integration must be changed according to the transformation.

These assumptions are written in the outline and transformation slides.

3. Transformation of One Random Variable

3.1 Problem Statement

Given:

$$Y = g(X)$$

Find:

$$f_Y(y)$$

3.2 Step-by-Step Derivation (Monotonically Increasing Case)

Step 1 — Start from definition of CDF

From the slide derivation:

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Since g is monotonic increasing,

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Thus,

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2 — Differentiate with respect to y

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Applying differentiation as shown in the slide:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

or equivalently,

.

$$f_Y(y) = f_X(x) \frac{dx}{dy} \quad \text{evaluated at } x = g^{-1}(y)$$

3.3 Monotonically Decreasing Case

From the derivation shown in the slide:

$$F_Y(y) = P(Y \leq y)$$

Since inequality reverses,

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Hence,

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

3.4 Final Result (As Shown in Slide Box)

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \quad \text{at } x = g^{-1}(y)$$

4. Worked Example — Transformation of One Random Variable

4.1 Given

From the example slide:

- $X \sim \text{Uniform}(-1, 1)$
- Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

PDF of X :

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

4.2 Step-by-Step Solution

Step 1 — Write transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Step 2 — Find inverse

From the slide:

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3 — Differentiate inverse

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 4 — Apply transformation formula

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Hence,

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

and zero otherwise.

5. Function of Two Random Variables

5.1 Problem Statement

Given:

$$Z = X + Y$$

Find:

$$f_Z(z)$$

The slide lists:

- Find PDF of Z
- Case when X and Y are independent
- Case when $X, Y \sim N(0, 1)$
- Case when exponential distributions with parameter λ are used

(Statements appear in the example slide.)

6. Step-by-Step Derivation for $Z = X + Y$

Step 1 — Start from definition of CDF

$$F_Z(z) = P(Z \leq z)$$

Substitute:

$$F_Z(z) = P(X + Y \leq z)$$

Step 2 — Express using joint PDF

From the graphical derivation shown in the slide (page showing shaded region):

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x,y) dx dy$$

The shaded region corresponds to all points satisfying $x + y \leq z$.

Step 3 — Integration Forms

The slide shows equivalent integration representations:

- Horizontal strip integration:

$$\int \left(\int f_{X,Y}(x,y) dx \right) dy$$

- Vertical strip integration:

$$\int \left(\int f_{X,Y}(x,y) dy \right) dx$$

Step 4 — Obtain PDF

Differentiate with respect to Z:

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

7. Summary of Results (From Lecture Slides)

1. Transformation uses:
 - CDF definition
 - inverse transformation
 - differentiation to obtain PDF.
2. Transformation formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

3. For functions of two random variables:
 - define CDF over transformed region,
 - integrate joint PDF over that region,
 - differentiate to obtain PDF.
-

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Lecture Scribe — Transformation of Random Variables

Course: CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Instructor: Dhaval Patel, PhD

Prepared strictly from lecture slides for exam revision

1. Scope and Objective of the Lecture

According to the outline slide:

1. Transformation of Random Variables
2. Function of Two Random Variables
3. Illustrative example: detailed derivation for the case

$$Z = X + Y$$

The stated objective in the lecture slides is:

- Learning transformation techniques for random variables.
 - Determining how to find the PDF of a new transformed random variable.
-

2. Definitions and Notation

2.1 Transformation of One Random Variable

Let:

- X be a random variable with PDF

$$f_X(x)$$

- A transformed random variable is defined as:

$$Y = g(X)$$

Definitions used in the derivation:

- CDF of Y :

$$F_Y(y) = P(Y \leq y)$$

- PDF of Y :

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

2.2 Function of Two Random Variables

Given two random variables:

$$Z = g(X, Y)$$

Illustrative example used in the lecture:

$$Z = X + Y$$

3. Assumptions and Conditions

From the lecture slides:

1. The PDF of the original random variable is known a priori.

2. The transformation $Y = g(X)$ is considered under monotonic conditions:
 - monotonically increasing
 - monotonically decreasing
3. For functions of two random variables:
 - the joint PDF $f_{X,Y}(x,y)$ is used.
4. Limits of the transformed variable must be adjusted according to the transformation.

These conditions appear explicitly in the transformation derivation slides.

4. Transformation of One Random Variable

4.1 Problem Statement

Given:

$$Y = g(X)$$

Find:

$$f_Y(y)$$

4.2 Step-by-Step Proof (Monotonically Increasing Case)

Step 1 — Start from definition of CDF

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Since $g(\cdot)$ is monotonic increasing:

$$P(g(X) \leq y) = P(X \leq g^{-1}(y))$$

Therefore,

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2 — Differentiate to obtain PDF

Differentiate with respect to y :

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Using differentiation as shown in the slide:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

or equivalently,

$$f_Y(y) = f_X(x) \frac{dx}{dy} \quad \text{at } x = g^{-1}(y)$$

4.3 Step-by-Step Proof (Monotonically Decreasing Case)

From the slide derivation:

$$F_Y(y) = P(Y \leq y)$$

Since inequality reverses:

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Thus,

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

4.4 Result (As Highlighted in the Slide)

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \quad \text{evaluated at } x = g^{-1}(y)$$

5. Worked Example — Transformation of One Random Variable

5.1 Given

From the example slide:

- $X \sim \text{Uniform}(-1, 1)$
- Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

PDF of X :

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

5.2 Step-by-Step Solution

Step 1 — Write transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Step 2 — Obtain inverse transformation

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3 — Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 4 — Apply transformation result

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Hence,

$$f_Y(y) = \frac{1}{\pi\sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

and zero otherwise.

All steps above follow the derivation written in the example slide.

6. Function of Two Random Variables

6.1 Problem Statement

Given:

$$Z = X + Y$$

Find:

$$f_Z(z)$$

The slide specifies the following cases:

1. Find PDF of Z
2. If X and Y are independent
3. If $X, Y \sim N(0, 1)$, prove $Z \sim N(0, 2)$
4. If X and Y are exponential with parameter λ , find $f_Z(z)$

(Statements are listed in the lecture slide.)

7. Step-by-Step Derivation for $Z = X + Y$

Step 1 — Definition of CDF

$$F_Z(z) = P(Z \leq z)$$

Substitute:

$$F_Z(z) = P(X + Y \leq z)$$

Step 2 — Express Using Joint PDF

From the diagram and derivation shown in the slide:

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x,y) dx dy$$

The shaded region in the diagram corresponds to the region satisfying $x + y \leq z$.

Step 3 — Integration Representation

The slide shows equivalent representations:

- Horizontal strip:

$$\int \left(\int f_{X,Y}(x,y) dx \right) dy$$

- Vertical strip:

$$\int \left(\int f_{X,Y}(x,y) dy \right) dx$$

Step 4 — Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

8. Summary of Results (From Lecture Slides)

1. Transformation of a random variable uses:

- CDF definition,
- inverse transformation,
- differentiation.

2. Transformation formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

3. For functions of two random variables:

- define CDF over transformed region,
- integrate joint PDF over that region,

- differentiate to obtain the PDF.

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Lecture Scribe — Transformation of Random Variables

Course: CSE400 – Fundamentals of Probability in Computing

Lecture 11: Transformation of Random Variables

Prepared strictly from lecture slides for exam preparation

1. Lecture Scope and Objective

From the lecture outline:

1. Transformation of Random Variables
2. Function of Two Random Variables
3. Illustrative Example: detailed derivation for the case

$$Z = X + Y$$

The objective stated in the lecture is:

- Learning transformation techniques for random variables.
 - Determining how to find the PDF of a transformed random variable.
-

2. Definitions and Notation

2.1 Transformation of One Random Variable

Let:

- X be a random variable with known PDF

$$f_X(x)$$

- A transformed random variable is defined as:

$$Y = g(X)$$

Definitions used in the derivations:

- CDF of Y :

$$F_Y(y) = P(Y \leq y)$$

- PDF of Y :

$$f_Y(y) = \frac{d}{dy} F_Y(y)$$

2.2 Function of Two Random Variables

Given two random variables:

$$Z = g(X, Y)$$

Illustrative example used in the lecture:

$$Z = X + Y$$

3. Assumptions and Conditions

From the lecture slides:

1. The PDF of the original random variable is known a priori.
2. The transformation $Y = g(X)$ is considered under monotonic conditions:
 - monotonically increasing
 - monotonically decreasing
3. For functions of two random variables:
 - joint PDF $f_{X,Y}(x,y)$ is used.
4. Limits must be changed appropriately after transformation.

These assumptions are indicated in the outline and transformation derivation slides.

4. Transformation of One Random Variable

4.1 Problem Statement

Given:

$$Y = g(X)$$

Find:

$$f_Y(y)$$

4.2 Step-by-Step Proof (Monotonically Increasing Case)

Step 1 — Definition of CDF

$$F_Y(y) = P(Y \leq y)$$

Substitute transformation:

$$F_Y(y) = P(g(X) \leq y)$$

Since $g(\cdot)$ is monotonic increasing:

$$F_Y(y) = P(X \leq g^{-1}(y))$$

Thus,

$$F_Y(y) = F_X(g^{-1}(y))$$

Step 2 — Differentiate with respect to y

$$f_Y(y) = \frac{d}{dy} F_X(g^{-1}(y))$$

Applying differentiation shown in the slide:

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y)$$

or equivalently,

$$f_Y(y) = f_X(x) \frac{dx}{dy} \quad \text{evaluated at } x = g^{-1}(y)$$

4.3 Step-by-Step Proof (Monotonically Decreasing Case)

From the derivation shown in the slide:

$$F_Y(y) = P(Y \leq y)$$

Since inequality reverses:

$$F_Y(y) = P(X \geq g^{-1}(y))$$

Hence,

$$F_Y(y) = 1 - F_X(g^{-1}(y))$$

Differentiating:

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

4.4 Result (As Presented in Slide)

$$f_Y(y) = \frac{f_X(x)}{\left| \frac{dy}{dx} \right|} \quad \text{evaluated at } x = g^{-1}(y)$$

5. Worked Example — Transformation of One Random Variable

5.1 Given

From the example slide:

- $X \sim \text{Uniform}(-1, 1)$
- Transformation:

$$Y = \sin\left(\frac{\pi X}{2}\right)$$

PDF of X :

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

5.2 Step-by-Step Solution

Step 1 — Write transformation

$$y = \sin\left(\frac{\pi x}{2}\right)$$

Step 2 — Inverse transformation

$$x = \frac{2}{\pi} \sin^{-1}(y)$$

Step 3 — Compute derivative

$$\frac{dx}{dy} = \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Step 4 — Apply transformation result

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

Substitute:

.

$$f_Y(y) = \frac{1}{2} \cdot \frac{2}{\pi} \frac{1}{\sqrt{1-y^2}}$$

Hence,

$$f_Y(y) = \frac{1}{\pi \sqrt{1-y^2}}$$

for

$$-1 < y < 1$$

and zero otherwise.

All intermediate steps are shown in the example slide.

6. Function of Two Random Variables

6.1 Problem Statement

Given:

$$Z = X + Y$$

Find:

$$f_Z(z)$$

The lecture slide lists the following cases:

1. Find the PDF of Z
2. If X and Y are independent
3. If $X, Y \sim N(0, 1)$, prove $Z \sim N(0, 2)$
4. If X and Y are exponential with parameter λ , find $f_Z(z)$

(Statements are listed in the slide.)

7. Step-by-Step Derivation for $Z = X + Y$

Step 1 — Definition of CDF

$$F_Z(z) = P(Z \leq z)$$

Substitute:

$$F_Z(z) = P(X + Y \leq z)$$

Step 2 — Express Using Joint PDF

From the diagram and derivation in the slide:

$$F_Z(z) = \iint_{x+y \leq z} f_{X,Y}(x,y) dx dy$$

The shaded region in the figure corresponds to the set of points satisfying $x + y \leq z$.

Step 3 — Integration Representation

The slide shows equivalent integration forms:

- Horizontal strip integration:

$$\int \left(\int f_{X,Y}(x,y) dx \right) dy$$

- Vertical strip integration:

$$\int \left(\int f_{X,Y}(x,y) dy \right) dx$$

Step 4 — Obtain PDF

$$f_Z(z) = \frac{d}{dz} F_Z(z)$$

8. Summary of Results (From Lecture Slides)

1. Transformation procedure:
 - express CDF,
 - apply inverse transformation,
 - differentiate to obtain PDF.
2. Transformation formula:

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{dx}{dy} \right|$$

3. For functions of two random variables:

- express CDF using joint distribution,
- integrate over transformed region,
- differentiate to obtain PDF.

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